

Feng PAN

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Research Interests

Statistical Physics, Tensor Network Algorithm, Machine Learning, AI for Quantum, Fault-tolerant Quantum Computation, Quantum Error Correction, Quantum Optimization.

Academic Positions

- Jan 2025 - **Assistant Professor**, *Science, Mathematics and Technology Cluster*, Singapore University of Technology and Design.
- Aug 2023 - **Postdoctoral Research Fellow**, *Centre for Quantum Technologies, National University of Singapore*,
Dec 2024 Advisor: Prof. Patrick Rebentrost.

Visiting Positions

- Feb 2023 - **Visiting Scholar**, *CAS Center For Excellence in Quantum Information and Quantum Physics, University of Science and Technology of China*,
Jul 2023 Host: Prof. Chaoyang Lu and Prof. Jianwei Pan.
- July 2022 - **Visiting Scholar**, *Institute of Theoretical Physics, Chinese Academy of Sciences*,
Jan 2023 Host: Prof. Pan Zhang.
- Sep 2015 - **Visiting Student**, *Massachusetts Institute of Technology, Cambridge*,
Jan 2016 Physics Department.

Education

- Sep 2017 - **Institute of Theoretical Physics, Chinese Academy of Sciences, Beijing**,
Jul 2022 *PhD in Theoretical Physics*, Supervisor: Prof. Pan Zhang.
- Sep 2012 - **Nanjing University of Aeronautics and Astronautics, Nanjing**,
Jun 2016 *Bachelor of Science, Mathematics & Physics*.

Funding

- 2026 **Advanced Quantum Algorithms and Solutions**,
Singapore Flagship Project under National Quantum Strategy, Pillar 4 Lead, 51 million SGD, 2026-2031.
- 2026 **Quantum-Enhanced Machine Learning Solvers for Optimisation Problems in Supply Chains and Logistics**,
MTC Young Individual Research Grant, 300K SGD, 2026-2029.
- 2025 **Tensor-Network solutions to large-scale combinatorial optimization problems**,
NQCH-NVIDIA collaboration under Quantum Engineering Program 3.0 sub-project, 697K SGD (including NVIDIA EOM), 2026-2027.
- 2025 **Towards implementing and decoding of logical quantum algorithms on present-day quantum hardware**,
MOE SKI Grant, 100K SGD, 2025-2028.

Awards

- 2023 **Second Prize, Beijing Natural Science Award,**
The highest scientific honor for researchers in Beijing; awarded annually to only 13 First Prize and 39 Second Prize recipients.
- 2023 **Outstanding Doctoral Dissertation Award, Beijing,**
Awarded to the top 0.1% of doctoral dissertations across Beijing.
- 2023 **Outstanding Doctoral Dissertation Award, Chinese Academy of Sciences,**
Awarded to the top 0.5% of doctoral dissertations at the Chinese Academy of Sciences.
- 2021 **National Scholarship for Doctoral Students, Ministry of Education of China,**
The highest national honor for doctoral students in China; awarded to the top 1%.
- 2021 **Principal Scholarship, Chinese Academy of Sciences,**
The highest honor for doctoral students within the Chinese Academy of Sciences; awarded to the top 1.25%.
- 2013 **National Scholarship for Bachelor Students, Ministry of Education of China,**
The highest national honor for undergraduate students in China; awarded to the top 1%.

Selected Talks

- Jul 2026 **Invited Talk at WAIC 2026**
World Artificial Intelligence Conference, Shanghai, China. [Event link.](#)
- May 2026 **Differentiable Maximum Likelihood Noise Estimation for Quantum Error Correction**
Invited talk at Google Quantum AI.
- Mar 2026 **Differentiable Maximum Likelihood Noise Estimation for Quantum Error Correction**
Invited talk at Quantinuum, London, UK.
- Dec 2025 **Tensor Networks as a Unified Language: From Quantum Computing to Artificial Intelligence**
Invited talk at Quantum Algorithms and Optimization in Athens, Athens, Greece [Talk link.](#)
- Oct 2025 **Machine Learning Solutions to Combinatorial Optimization Problems: Insights from Statistical Physics**
Invited talk at The 16th Asia Pacific Physics Conference, Haikou, China.
- Aug 2025 **Advancements in Decoding for Quantum Error Correction**
Invited talk at The Conference of Condensed Matter Physics, Liyang, China.
- Feb 2024 **Arbitrary Tensor Network Algorithm: Theory, Methods and Applications**
Invited talk at IPAM workshop TNK 2024, UCLA, Los Angeles, USA, [Talk link.](#)
- Oct 2023 **Classical Simulation of Quantum Circuits by Arbitrary Tensor Network Algorithms**
Prof. Jens Eisert Group, Online.
- Sep 2023 **Classical Simulation of Quantum Circuits by Arbitrary Tensor Network Algorithms**
Centre for Quantum Technologies, Singapore.
- Nov 2021 **Contracting Arbitrary Tensor Networks: Approximate and Exact Approach with Applications in Graphical Models and Quantum Circuit Simulations**
Perimeter Institute, Waterloo, Canada, [Talk link.](#)

Service

- PC Member [QIP 2027](#), AQIS 2026
- Journal Referee Physical Review Letters, Physical Review A, Physical Review B, Physical Review Research, PRX Quantum, npj Quantum Information, Quantum, Neural Computing
- Conference Referee Conference on the Theory of Quantum Computation, Communication and Cryptography (TQC)

Patents

- [1] Pan Zhang, **Feng Pan**, Methods, devices and quantum virtual machines for simulating probabilistic amplitudes of quantum states, CN114254755, 2022.

Publications

I have authored 12 journal articles and 1 conference proceeding, in addition to 10 preprint papers. Notably, 5 of these articles have been published in **Physical Review Letters**, the leading journal in the field of Physics, and one in **Nature Computational Science**. Detailed information about my publications can be found on [Google Scholar](#).

- [1] **Feng Pan**, and Pan Zhang, *Simulation of Quantum Circuits Using the Big-Batch Tensor Network Method*, *Phys. Rev. Lett.* **128**, 030501 (2022) (Editors' Suggestion).
Blog discussions: [Scott Aaronson](#), [Gil Kalai](#).
- [2] **Feng Pan**, Keyang Chen, and Pan Zhang, *Solving the Sampling Problem of the Sycamore Quantum Circuits*, *Phys. Rev. Lett.* **129**, 090502 (2022) (Editors' Suggestion).
Media highlights for item [2]: [Science](#), [The Quantum Insider](#), [New Scientist](#).
- [3] **Feng Pan**, Pengfei Zhou, Sujie Li, and Pan Zhang, *Contracting Arbitrary Tensor Networks: General Approximate Algorithm and Applications in Graphical Models and Quantum Circuit Simulations*, *Phys. Rev. Lett.* **125**, 060503 (2020).
- [4] **Feng Pan**, Hanfeng Gu, Lvlin Kuang, Bing Liu, and Pan Zhang, *Efficient Quantum Circuit Simulation by Tensor Network Methods on Modern GPUs*, *ACM Transactions on Quantum Computing* **5**, 1-26 (2024).
- [5] Zi-Song Shen, **Feng Pan**, Yao Wang, Yi-Ding Men, Wen-Biao Xu, Man-Hong Yung, and Pan Zhang, *Free-Energy Machine for Combinatorial Optimization*, *Nature Computational Science* **5**, 322-332 (2025).
- [6] Sujie Li, **Feng Pan**, Pengfei Zhou, and Pan Zhang, *Boltzmann Machines as Two-Dimensional Tensor Networks*, *Phys. Rev. B* **104**, 075154 (2021).
- [7] Yijia Wang, Yuwen Ebony Zhang, **Feng Pan**, and Pan Zhang, *Tensor Network Message Passing*, *Phys. Rev. Lett.* **132**, 117401 (2024) (Editors' Suggestion).
- [8] **Feng Pan**, Pengfei Zhou, Hai-Jun Zhou, and Pan Zhang, *Solving Statistical Mechanics on Sparse Graphs with Feedback-Set Variational Autoregressive Networks*, *Phys. Rev. E* **103**, 012103 (2021).
- [9] Hanyan Cao, **Feng Pan**, Yijia Wang, and Pan Zhang, *qecGPT: Decoding Quantum Error-Correcting Codes with Generative Pre-Trained Transformers*, [arXiv:2307.09025](#).
- [10] Xian-He Zhao, Han-Sen Zhong, **Feng Pan**, et al., *Leapfrogging Sycamore: Harnessing 1432 GPUs for 7x Faster Quantum Random Circuit Sampling*, *National Science Review* **12**, nwae317 (2025).
- [11] Ya-Peng Hu, **Feng Pan**, and Xin-Meng Wu, *The Effects of Massive Graviton on the Equilibrium Between the Black Hole and Radiation Gas in an Isolated Box*, *Phys. Lett. B* **772**, 553-558 (2017).
- [12] Hanyan Cao, Shoukuan Zhao, Dongyang Feng, Zisong Shen, Haisheng Yan, Tang Su, Weijie Sun, Huikai Xu, **Feng Pan**, Haifeng Yu, and Pan Zhang, *Exact Decoding of Quantum Error-Correcting Codes*, *Phys. Rev. Lett.* **134**, 190603 (2025) (Editors' Suggestion).
- [13] Rong Fu, Zhongling Su, Han-Sen Zhong, Xiti Zhao, Jianyang Zhang, **Feng Pan**, et al., *Surpassing Sycamore: Achieving Energetic Superiority Through System-Level Circuit Simulation*, *SC24: International Conference for High Performance Computing, Networking, Storage and Analysis* (2024).
- [14] Hanyan Cao, **Feng Pan**, Dongyang Feng, Yijia Wang, and Pan Zhang, *Generative Decoding for Quantum Error-Correcting Codes*, [arXiv:2503.21374](#).
- [15] Naixu Guo, **Feng Pan**, and Patrick Rebentrost, *Estimating Properties of a Quantum State by Importance-Sampled Operator Shadows*, [arXiv:2305.09374](#).
- [16] Hanyan Cao, Ge Yan, Yuxuan Du, and **Feng Pan**, *Maximum Likelihood Decoding of Quantum Error Correction Codes*, [arXiv:2605.17230](#).
- [17] Hanyan Cao, Dongyang Feng, Cheng Ye, and **Feng Pan**, *Differentiable Maximum Likelihood Noise Estimation for Quantum Error Correction*, [arXiv:2602.19722](#).
- [18] Ge Yan, Shanchuan Li, Shiyi Xiao, Pengyue Ma, Hanyan Cao, **Feng Pan**, and Yuxuan Du, *Efficient Foundation Decoders for Fault-Tolerant Quantum Computing*, [arXiv:2606.27119](#).
- [19] **Feng Pan**, Hanfeng Gu, Paul Springer, and Xipeng Li, *Parallelizing Large-Scale Tensor Network Contraction on Multiple GPUs*, [arXiv:2606.01852](#).
- [20] Yijia Wang, Xuanzhao Gao, Pan Zhang, **Feng Pan**, and Jinguo Liu, *Branch-and-Bound Tensor Networks for Exact Ground-State Characterization*, [arXiv:2602.05470](#).

- [21] Hanyan Cao, Yijia Wang, **Feng Pan**, and Pan Zhang, *Integrating Neural Networks and Tensor Networks for Computing Free Energy*, [Communications in Theoretical Physics](#) **77**, 095602 (2025).
- [22] Hidetaka Manabe, Hanfeng Gu, and **Feng Pan**, *Classical Simulation and Design Frontiers for IBM's Doped Clifford Sampling Experiment*, [arXiv:2608.13110](#).
- [23] Simon Martiel, Jay-U Chung, Alireza Seif, Soumik Ghosh, Ian Hincks, Abhinav Deshpande, Hidetaka Manabe, Hanfeng Gu, **Feng Pan**, Bill Fefferman, Jay M. Gambetta, and Ali Javadi-Abhari, *Sampling Hard Circuits with Verifiably High Fidelity*, [arXiv:2607.25941](#).